INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME

CG

INDEX NUMBER

BIOLOGY

8876/02

Paper 2 Structured and Free-response Questions

28 August 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, CG and index number on page 1, 7 and 11 in the spaces provided.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 10
2	/ 14
3	/ 13
4	/ 8
5 or 6	/ 15
Total	/ 60

This document consists of **16** printed pages.

Section A

Answer **all** the questions in this section.

1 Fig. 1.1 shows part of a eukaryotic cell.

Fig. 1.1

(a) **(i)** Outline the role of organelle labelled **A**.

[3]

(ii) Identify **one** molecule with different modes of transport across the membrane of **A** and explain their modes of transport.

[2]

(b) **(i)** Explain the significance of glycolysis in aerobic respiration.

....

....

....

....

....

.... [3]

An experiment was carried out to investigate the effect of temperature on respiration in isolated mitochondria extracted from a worm. Respiratory substrate was provided and oxygen consumption was monitored at 15°C, 25°C and 35°C.

Fig. 1.2 shows the temperature coefficients, Q_{10}, when temperature is increased from 15°C to 25°C and from 25°C to 35°C.

Q_{10} measures the ratio of the rate of respiration when the temperature increases by 10°C.

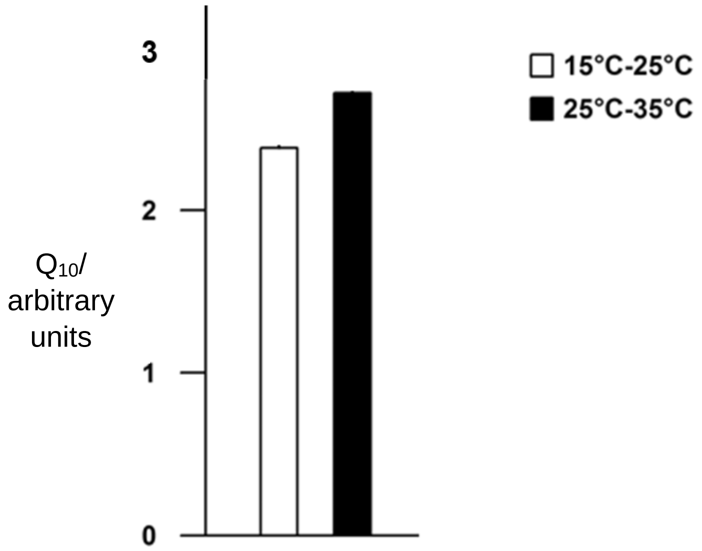

Fig. 1.2

(ii) Describe and explain the effect of temperature on the Q_{10} of mitochondria respiration.

....

....

....

.... [2]

[Total: 10]

2 Fig. 2.1 is a photomicrograph of plant root cells near the growing top. Some of the cells are undergoing mitosis.

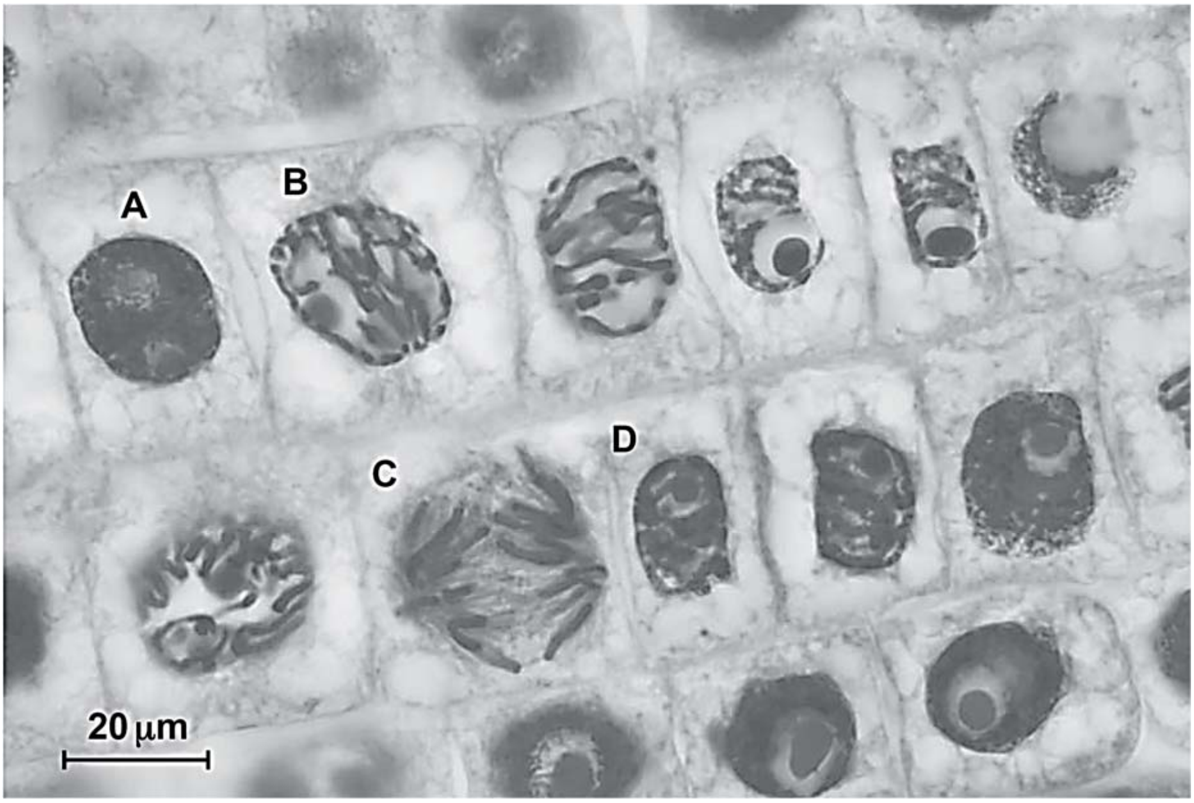

Fig. 2.1

(a) State **one** feature, visible in Fig. 2.1, which indicates that the section is taken from a plant tissue and not animal tissue.

.... [1]

(b) State the letter, **A** to **D**, of the cell in Fig. 2.1 which is in:

(i) prophase ....

(ii) anaphase .... [2]

(c) Explain why the growth of roots involves mitosis and not meiosis.

....

....

....

....

....

.... [3]

(d) Fig. 2.2 represents one complete cell cycle for the plant root cells.

(i) Complete Fig. 2.2 by naming the stages represented by **J**, **K** and **L**.

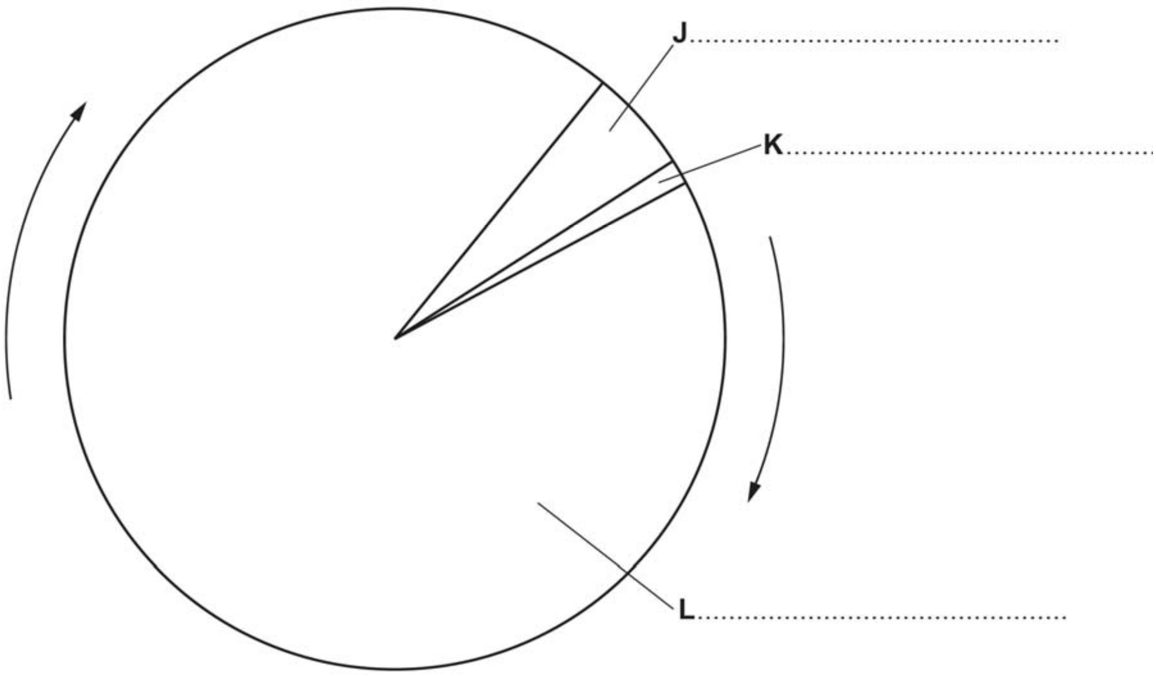

Fig. 2.2 [3]

(ii) Some bacteria infect the plant to result in tumour formation at the infected tissues as shown in Fig. 2.3.

Research on the tumour formation processes revealed that there was an unusual excessive production of auxin, an important plant growth hormone. Auxin has been found to regulate gene expression resulting in cell division, expansion, differentiation and specialisation.

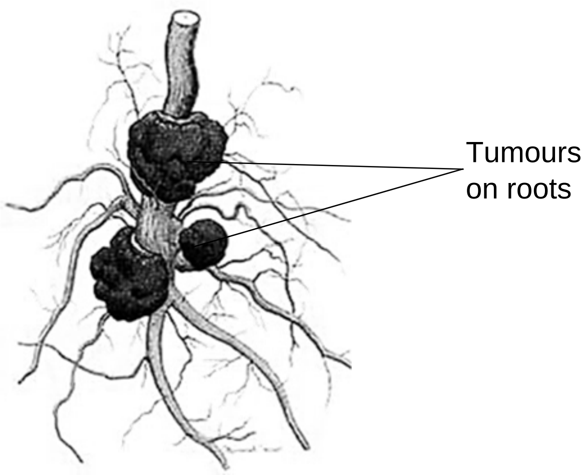

Fig. 2.3

Suggest how the bacteria infection resulted in tumour formation.

[3]

(iii) Suggest why plants with these tumours if left untreated will eventually die.

[2]

[Total: 14]

Name: ____________________

Index No.: ___________ CG: 17____

3 In sickle cell anaemia the recessive allele Hb^S replaces the normal allele Hb^A.

- The frequency of Hb^S is much higher in West Africa than in most parts of the world.
- The frequency of Hb^S corresponds with the distribution of malaria.

(a) Explain what is meant by the term *allele*.

....

.... [1]

(b) State whether the likely life expectancy is high or low in West Africa for individuals with the following genotypes. In each case give a reason for your answer.

Hb^AHb^A ....

.... [1]

Hb^AHb^S ....

.... [1]

Hb^SHb^S ....

.... [1]

(c) Suggest why the frequency of Hb^S allele corresponds to the distribution of malaria.

....

....

....

....

....

.... [3]

(d) Describe how the Hb^S allele came about.

....

....

....

.... [2]

(e) A man with sickle-cell anaemia marries a woman without sickle-cell anaemia whose mother also has sickle-cell anaemia. Using a genetic diagram, find the probability that the couple would give birth to a son with sickle-cell anaemia.

[4]

[Total: 13]

4 Reef-building corals are marine invertebrates found in shallow, clear, tropical oceans. The corals secrete an exoskeleton of calcium carbonate that becomes the underlying structure of the coral reef ecosystem.

Zooxanthellae are a group of unicellular algae from the genus *Symbiodinium* that live within the cells of reef-building corals. The relationship has been described as mutualistic since it is beneficial to both coral and zooxanthellae.

(a) Evidence shows that the mutualistic relationship between zooxanthellae and reef-building corals has evolved by free-living algae invading corals that did not contain algae.

(i) Corals that do not need zooxanthellae can live at a greater depth than reef-building corals.

Explain why this is so.

....

....

....

.... [2]

(ii) Suggest **two** benefits **to the zooxanthellae** of their association with the corals.

....

....

....

....

....

.... [2]

(b) Under conditions of stress the relationship between the reef-building corals and the zooxanthellae can break down. Loss of zooxanthellae and the subsequent whitening that occurs, shown in Fig. 4.1, is known as coral bleaching. Coral bleaching can lead to death of the coral.

Fig. 4.1

(i) Suggest **two** reasons why permanent loss of zooxanthellae can lead to the death of the coral.

....

....

....

.... [2]

(ii) Increased sea temperature associated with global climate change is known to be an environmental stress that can cause coral bleaching. The temperature range for healthy survival of reef-building coral is 25°C to 29°C.

Explain why the areas of sea containing coral reefs are susceptible to increased temperature resulting from global climate change.

....

....

....

.... [2]

[Total: 8]

Name: ___________________

Index No.: __________ CG: 17____

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a), (b) a**s indicated in the question.

Either

5 **(a)** Stem cells research offers a great number of potential benefits to humans and has helped scientists to understand many cellular processes naturally involved in the growth and development of organisms.

Describe the role of stem cells in humans and discuss the ethical issues that arise from stem cells research. [8]

(b) Outline how genes found on the DNA are expressed to result in the phenotype of an organism. [7]

[Total: 15]

Or

6 **(a)** Proteins play important roles in the metabolic processes of both plants and animals. In particular, enzymes assist in the catalysis of key activities in the production of organic matter in plants using carbon dioxide as its raw material.

Describe the roles of enzymes in photosynthesis. [9]

(b) Membranes of organelles in photosynthesis are key in the light dependent stage of photosynthesis. Metabolic poisons are known to disrupt key events in the light dependent stage.

Explain the possible ways metabolic poisons can inhibit light dependent reaction in photosynthesis. [6]

[Total: 15]